

21	The electric field strength due to a point charge at a distance r from the charge is E. What is the electric field strength at distance $2r$ from the point charge?							
	A	$\frac{E}{4}$	B	$\frac{E}{2}$	C	$2E$	D	$4E$
	Answer: A $E \propto \frac{1}{r^2}, \text{ so } \frac{E}{E'} = \frac{r'^2}{r^2} \rightarrow \frac{E}{E'} = \frac{4r^2}{r^2}$ Hence $E' = \frac{E}{4}$							

22	A wedge shaped metal conductor of length l , varying width and uniform thickness is
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